



B9AI103 Recommender Systems

CA ONE

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Overall marks will be awarded according to the following criteria:	3
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Assessment:

Item	Description
Module Title:	Recommender Systems
Module Code:	B9AI103
Module Lecturer (s):	Nitya Govindaraju
Programme/Cohort:	MSc Artificial Intelligence
Method of Assessment:	CA ONE
Percentage (%) Weighting:	60%
MIMLOs being assessed:	1, 2, 5, 6
Assessment Number:	ONE
Individual/Group:	Group (3-4 participants per group)
Issue Date:	Week 2
Submission Date:	Week 7
Feedback Date:	Week 9
Feedback Strategy:	Moodle

Assignment Task

Group Project on Model Development and Evaluation Techniques

Students will work in groups to design, develop, and evaluate a recommender system for a specific use case (e.g., e-commerce, media, job recommendations).

Overall marks will be awarded according to the following criteria:

Task	Description	Marks	MIMLO Alignment
Part 1: System Design and Core Implementation	Select a real-world recommendation problem and dataset. Design and implement at least two distinct types of recommender systems (e.g., Collaborative Filtering and Content-Based Filtering or Hybrid). Document the rationale for the chosen models, data pre-processing, and feature engineering.	35	1, 5
Part 2: Evaluation and Cold-Start Mitigation	Apply and report on a comprehensive offline evaluation strategy using appropriate metrics like NDCG and MAP. Implement and discuss a strategy to address a key challenge, such as the Cold Start problem or Scalability, demonstrating its effectiveness.	40	2, 5
Part 3: Generative AI Analysis & Presentation	Research and discuss the role of Generative AI (e.g., GANs, VAEs) in enhancing recommender systems. Compare and contrast how a generative approach could potentially enhance your implemented systems versus the traditional methods used. Present the project's design, implementation, evaluation, and Generative AI analysis to the class (Group Presentation).	25	6
Total		100	1, 2, 5, 6

Deliverables:

- **Group Report (Max 3,000 words, excluding code/appendices):** A formal report detailing the entire process, including system design, algorithm choice, implementation steps, evaluation results, and the Generative AI analysis. **(NEEDS TO BE SUBMITTED as separate PDF file; DO NOT ZIP this file)**
- **Code Repository (GitHub/similar):** Submission of all source code, clearly documented and functional. OR **Python Notebooks**
- **Group Presentation (10-15 minutes):** A presentation summarizing the project and key findings.

Resources

Software or tools required/ useful:

- Programming Languages: Python
- Libraries/Frameworks: TensorFlow, PyTorch, Scikit-learn, Surprise, SHAP, LIME, Pandas, NumPy
- Data Visualization: Matplotlib, Seaborn
- Collaboration Tools: GitHub, Jupyter Notebook

Templates, files, data tables provided by the lecturer:

- Project template for documenting group projects
- Sample datasets for training and evaluation
- Evaluation metrics sheet

Highlight the Appropriate Selection as per the programme Module and Assessment Document

1	2	3	4	5
NO AI	AI-ASSISTED IDEA GENERATION AND STRUCTURING	AI-ASSISTED EDITING	AI TASK COMPLETION, HUMAN EVALUATION	FULL AI
The assessment is completed entirely without AI assistance. This level ensures that students rely solely on their knowledge, understanding, and skills. AI must not be used at any point during the assessment.	AI can be used in the assessment for brainstorming, creating structures, and generating ideas for improving work. No AI content is allowed in the final submission.	AI can be used to make improvements to the clarity or quality of student created work to improve the final output, but no new content can be created using AI. AI can be used, but your original work with no AI content must be provided in an appendix.	AI is used to complete certain elements of the task, with students providing discussion or commentary on the AI-generated content. This level requires critical engagement with AI generated content and evaluating its output. You will use AI to complete specified tasks in your assessment. Any AI created content must be cited.	AI should be used as a 'co-pilot' in order to meet the requirements of the assessment, allowing for a collaborative approach with AI and enhancing creativity. You may use AI throughout your assessment to support your own work and do not have to specify which content is AI generated

Grading Criteria		Evaluative Criteria/Performance Level Descriptors (what does performance at this level look like?)		
Scale	Grade Band	Criterion 1: System Design & Implementation (MIMLOs 1, 5)	Criterion 2: Evaluation & Challenge Mitigation (MIMLOs 2, 5)	Criterion 3: Generative AI Analysis & Presentation (MIMLO 6)
Exceptional	90-100%	Demonstrates original design and masterful implementation of two complex, appropriate systems. Code is highly efficient and robust.	Evaluation is comprehensive, using advanced metrics. Mitigation of the challenge (e.g., cold start) is novel, fully implemented, and results are convincingly validated.	The Generative AI analysis is highly insightful, critically evaluating trade-offs and proposing future work with sophistication. Presentation is compelling and professionally delivered.
Outstanding	80-89%	Excellent selection and technically flawless implementation of two distinct recommender systems. Strong justification for design choices and data handling.	Highly effective offline evaluation strategy is applied with accurate metrics. Excellent, well-implemented solution to the identified challenge, with clear results.	Detailed and critical analysis of GenAI's role in the system. Comparison to traditional methods is sharp and well-supported. Presentation is clear and engaging.
Excellent	70-79%	Solid selection and competent, correct implementation of two distinct systems. Demonstrates a deep understanding of core recommendation architectures.	Effective evaluation methodology is applied using appropriate metrics (NDCG, MAP). A clear, workable solution to the identified challenge is implemented and documented.	Good understanding of GenAI's application in the context of the project. Comparison to traditional methods is logical and well-structured. Presentation is professional.
Very Good	60-69%	Implementation of two distinct systems is mostly correct, demonstrating a very good understanding of the module content. Minor technical or design flaws may exist.	Evaluation is generally correct and uses standard metrics. The mitigation strategy for the challenge is plausible and partially implemented.	Clearly discusses the potential of GenAI in enhancing the system. The presentation is well-organized and covers all required areas.

Good	50-59%	Correct implementation of the two required systems. Demonstrates a good conceptual understanding but implementation may lack optimization or clarity.	Basic evaluation strategy is applied, but may lack depth (e.g., relying heavily on simple metrics). The challenge is identified and a theoretical solution is discussed, but implementation is weak.	Basic discussion of GenAI is included, showing foundational awareness of its role in recommenders. Presentation is adequate.
Sufficient	40-49%	Demonstrates a foundational understanding through a minimally functional implementation of the two required systems. Contains notable technical errors or poor design choices.	Attempts to apply evaluation, but results or metric selection may be flawed or incomplete. The challenge is identified but the proposed solution is poorly articulated or missing.	Mentions GenAI but the discussion is superficial or lacks specific relevance to the project. Presentation is weak.
Unfulfilled	0-39%	Significant technical errors, one or both systems are non-functional, or implementation shows a lack of foundational understanding.	Evaluation is missing or completely incorrect. No attempt to address the challenge is evident.	Lacks a discussion of Generative AI and/or presentation is missing or severely deficient.
MIMLO Alignment (see Module Guide for Minimum Intended Learning Outcomes for Module)				
MIMLO to which criterion is aligned:	1,5	2,5	6	

General Requirements for Students:

1. All assignments must be submitted no later than the stated deadline (date and time).
2. Assignments submitted after the latest deadline specified (including any approved extension deadline) are considered late and penalised according to the Quality Assurance Handbook (QAH) Part B Section 5.2.2.6 as follows:
 - a. A penalty of 2 marks will be applied per day or part thereof (including weekends and public holidays) for an ongoing failure to submit beyond the submission deadline.
 - b. An examiner has the right to refuse to mark the assignment if the submission instructions have not been observed.

- c. Where a late assessment is submitted within 14 days of the deadline, and is of a passing standard, the late penalty is capped (such that the minimum grade that can be awarded is 40% for the late submission).
 - d. Where a late assessment is submitted more than 14 days after the deadline, it will receive 0%. The lecturer may, at their discretion, review the submission for feedback.
 - e. Where the assessment is undertaken in a group, the piece of work should be submitted in its complete entirety, and any penalty for late submission incurred applies to all group members.
3. Extensions to assignment submission deadlines will not be granted, other than in exceptional circumstances. To apply for an extension please go to <https://students.dbs.ie/dashboard/SCCM> and open a ticket.
4. All relevant provisions of the Assessment Regulations must be complied with (see [QAH B.5](#)).
- a. Students are required to refer to the assessment regulations in their Programme Handbook, and on the [Student Website](#).
 - b. Dublin Business School penalises students who engage in academic impropriety (i.e. plagiarism, collusion and/or copying, ghost writing/ essay mills, improper use of Generative Artificial Intelligence software).
 - i. Refer to the College's [Generative AI Guidelines HERE](#) for further information.
 - c. Guides on referencing are available on the Library website: <https://libguides.dbs.ie/referencing>
 - d. Text-matching analysis software is integrated in Moodle to generate a report regarding the degree of text-matching in a submission.
5. Students are required to retain a copy of each assignment submitted, until the issuing of a transcript indicating the mark awarded and the closure of the Appeal period (2 weeks following the release of final results).
- a. Results can only be appealed following the release of final results, and the Appeal form must be submitted to the Exams Office within the Appeal period.
 - b. An appeal must be based on valid grounds (see the Appeals Policy [QAH B.3.5](#)), dissatisfaction with a grade is not sufficient grounds for an appeal.
 - c. Assignments must be appropriately packaged and presented.
 - d. All assignments should be submitted to your subject/course page on Moodle by the deadline date.
 - e. Where a submission involves digital media (i.e formats other than Word, Powerpoint or PDF), it is the submitting students' responsibility to ensure the media is appropriately labelled, fully working and they must retain a copy.
 - f. Components of an assessment which are not included in the final submission cannot normally be subsequently accepted for grading. It is the student's responsibility to ensure their file is uploaded correctly.
 - g. Include an electronic **cover sheet** with the following details to the front of the assignment (see below)

6. Assignments that *breach* the word count requirements will be penalised. *There is a 10% discretion, either way, applicable in terms of word count.*
7. When you submit your assignment you will be asked to click on a button which will declare the following:

By ticking this box I am confirming that this assignment/exam is all my own work. Any sources used have been referenced.

I have read the College rules regarding plagiarism in the QAH Part B Section 3 and understand that penalties will be applied accordingly if work is found not to be my own. All work uploaded is submitted via Ouriginal, whereby a text-matching report will show any similarities with other texts.